



Jordi Cerdà-Bautista

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Research Interests

Causal inference and machine learning applied to food security, climate variability, conflict, and humanitarian policy in low- and middle-income countries. Methodological focus on heterogeneous treatment effect estimation, causal discovery, and spatial/panel methods. Substantive focus on sub-Saharan Africa, with operational links to WFP, JRC, and FAO humanitarian datasets.

Education

Universitat de València

PhD Candidate, Image Processing Laboratory (IPL)

Thesis: *Causal Machine Learning for Food Security: Evidence from Socio-Environmental Shocks*

Supervisors: Gustau Camps-Valls & Vasileios Sitokostantinou

Defense expected: early 2027

València, Spain

Sep. 2022–Present

Universitat de València

M.Sc. in Data Science

Thesis: *Deep Learning Methods for Atmospheric Correction of Satellite Images*

Supervisors: Luis Gómez Chova & Gonzalo Mateo García

València, Spain

2020–2022

Università degli Studi di Torino

Erasmus Programme

Turin, Italy

2018–2019

Universitat de València

B.Sc. in Physics

Thesis: *Interference with a spatially incoherent polarized source*

Supervisors: Genaro Saavedra Tortosa & Juan Carlos Barreiro Hervás

València, Spain

2015–2020

Honors & Awards

WCRP Open Science Conference

Best In-Person Poster Award

Poster: *Understanding food insecurity in Africa through data-driven causal inference methods.*

Kigali, Rwanda

Oct. 2023

Experience

Image Processing Laboratory (IPL), Universitat de València

Doctoral Researcher

Causal machine learning for food security across sub-Saharan Africa: humanitarian intervention evaluation, climate–price interaction analysis, conflict spillover effects, and micronutrient pathway decomposition. Co-developer of *CauseMe*, a web platform for causal discovery in Earth and climate sciences.

València, Spain

Feb. 2022–Present

Capgemini S.A.

Data Scientist Intern

ML models, SQL, web design.

València, Spain

Oct. 2021–Feb. 2022

Publications

Peer-Reviewed Journal Articles.....

2024: J. M. Tárraga, E. Sevillano-Marco, J. Muñoz-Marí, M. Piles, V. Sitokonstantinou, M. Ronco, M. T. Miranda, **J. Cerdà**, G. Camps-Valls. *Causal discovery reveals complex patterns of drought-induced displacement*. **iScience** 27(9), 2024. [Cell Press; JCR Q1]

Manuscripts in Preparation.....

2026: **J. Cerdà-Bautista**, V. Sitokonstantinou, J.M. Veiga López-Peña, D. Piovani, J.M. Tárraga, G. Camps-Valls. *Credit Access is Associated with Improved Food Security in the Horn of Africa*. Manuscript under revision.

2026: **J. Cerdà-Bautista**, V. Sitokonstantinou, H. Durand, G. Varando, M. Ronco, G. Camps-Valls. *Climate Variability Amplifies Food Insecurity During Price Spikes*. Manuscript under revision.

2026: **J. Cerdà-Bautista**, V. Voukelatou, V. Sitokonstantinou, D. Piovani, G. Camps-Valls. *Causal Determinants of Micronutrient Deficiency across LMIC settings*. Manuscript in preparation.

2026: **J. Cerdà-Bautista**, V. Sitokonstantinou, G. Camps-Valls. *Conflict Lethality Drives Food Insecurity in Sub-Saharan Africa*. Manuscript in preparation; preprint forthcoming (Summer 2026).

Preprints.....

2024: V. Sitokonstantinou, E. D. S. Porras, **J. Cerdà-Bautista**, M. Piles, I. Athanasiadis, G. Camps-Valls. *Causal Machine Learning for Sustainable Agroecosystems*. arXiv:2408.13155.

Peer-Reviewed Conference Proceedings.....

2024: **J. Cerdà-Bautista**, J. M. Tárraga, V. Sitokonstantinou, G. Camps-Valls. *Assessing the Causal Impact of Humanitarian Aid on Food Security*. IGARSS 2024 — IEEE International Geoscience and Remote Sensing Symposium, Athens.

Software

CauseMe: Web platform for causal discovery in Earth and climate sciences. Implements PCMCi, LinearVAR, Granger2D, and GraphEM with interactive edge-bundling visualisation and an LLM-based result-summarisation module. causeme.uv.es

Causal4FS: Open-source pipelines for causal effect estimation on humanitarian and food-security data — covering cash-based transfers, price spikes, conflict escalations, and micronutrient pathways. github.com/jordicbau

Funded Research Projects

2024–Present: *ThinkingEarth* — EU Horizon Europe Research & Innovation Programme. Copernicus Foundation Models for socio-environmental applications.

2023–2024: *Causal4Africa* — Microsoft Climate Research Initiative. Causal evaluation of humanitarian interventions in Africa.

2022–2023: *SCALE: Causal Inference in the Human–Biosphere Coupled System* — Fundación BBVA.

2020–2022: *DeepCube* — EU Horizon 2020 (contributor). Explainable AI for Copernicus Earth Observation data.

Selected Presentations

Dec. 2025: Poster: *Causal Effects of Price Spikes on Food Insecurity*. EurlIPS Workshop on Causality for Impact, Copenhagen.

Jun. 2025: Talk: *Human and Environmental Causal Effects on Food Security in Africa*. ESA Living Planet Symposium, Vienna.

Apr. 2025: Talk: *Hierarchical Causal Graph-Based Methods for Imputing Food Insecurity*. EGU General Assembly, Vienna.

Jul. 2024: Talk: *Assessing the Causal Impact of Humanitarian Aid on Food Security*. IGARSS, Athens.

May 2024: Poster: *Evaluating the Causal Impact of Humanitarian Interventions on Food Insecurity in Climate-*

Vulnerable Regions of Africa. ESA EO4AGRI Workshop, Frascati.

Apr. 2024: Talk: *Causal Evaluation of Humanitarian Aid on Food Security*. EGU General Assembly, Vienna.

Oct. 2023: Poster (best in-person award): *Understanding Food Insecurity in Africa Through Data-Driven Causal Inference Methods*. WCRP Open Science Conference, Kigali.

Apr. 2023: Talk: *Causal Inference to Study Food Insecurity in Africa*. EGU General Assembly, Vienna.

Additional schools and workshops attended: Nordic Probabilistic AI School (NTNU, Trondheim, 2023); *When Causal Inference meets Statistical Analysis* (CNAM, Paris, 2023).

Technical Skills

Methods: Causal effect estimation; causal discovery; panel and spatial inference; refutation and sensitivity testing.

Programming: Python (primary), R, SQL, MATLAB.

ML & Causal Libraries: EconML, DoWhy, Tigramite, DoubleML, scikit-learn, TensorFlow, PyTorch, XGBoost.

Geospatial & EO: Google Earth Engine, xarray, rasterio, GeoPandas; NDVI / SPI / FPAR pipelines.

Engineering & Web: Git, PySpark, R Shiny, Dash, Flask, HTML/CSS/JavaScript.

Other: $\text{\LaTeX}$, Microsoft Office.

Selected Training

Universitat de València: Causal Discovery and Inference; Kernel Methods in Machine Learning; Deep Learning for Earth Sciences.

Coursera & Pluralsight: Machine Learning with TensorFlow on Google Cloud (specialisation); Advanced Machine Learning on Google Cloud (specialisation); Apache Spark fundamentals; Google Cloud core infrastructure.

Languages

Catalan: Native

Spanish: Native

English: Fluent

Italian: Intermediate

References

Available on request.